



# Om Sharma

Roll No.: 250103069

B.Tech - Mechanical Engineering

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## EDUCATION

| Degree/Certificate     | Institute/Board                          | CGPA/Percentage | Year         |
|------------------------|------------------------------------------|-----------------|--------------|
| B.Tech                 | Indian Institute of Technology, Guwahati | 8.30 (Current)  | 2025-Present |
| Senior Secondary (XII) | CBSE Board                               | 86.2%           | 2025         |
| Secondary (X)          | CBSE Board                               | 93.6%           | 2023         |

## PROJECTS

- GradeOps: AI-Powered Human-in-the-Loop Exam Grading Platform** Mar. 2025 - Present  
*Coding Club, IIT Guwahati · Python, FastAPI, React, LangGraph, Gemini, Nemotron, PostgreSQL* Github
  - Built a platform that **OCRs handwritten exam scripts via NVIDIA Nemotron**, scores each answer against instructor rubrics using **Google Gemini**, and routes every result through a **human review queue** before grades are released.
  - Engineered a **LangGraph state machine** orchestrating the two-stage AI pipeline with **SQLite checkpointing** so pending reviews survive server restarts; backend served via **FastAPI on PostgreSQL** with role-based cookie auth.
  - Developed the **React frontend from scratch** – Instructor and TA dashboards, keyboard-friendly review queue, and a premium animated landing page with **Three.js** visuals.
- Delhivery Network Intelligence: Graph-Based Logistics Optimization** Jun. 2025  
*Consulting & Analytics Club, IIT Guwahati · Python, XGBoost, NetworkX, node2vec, Pandas, Streamlit* Github
  - Modelled Delhivery's logistics network as a directed weighted graph on 144,900 scans to identify that 5 hubs account for 30.8% of total network excess delay; top bottleneck carries 12.5% of network delay alone.
  - Built graph-enhanced XGBoost ETA model using node2vec hub embeddings and betweenness centrality features, achieving **MAE 28 min vs OSRM's 108 min (74% improvement)** and 58.3% on-target accuracy vs 4.1%.
  - Delivered FTL vs Carting decision framework and upgrade impact analysis estimating **28% reduction in late deliveries** and **~Rs. 1.5 cr/yr** recovered revenue from top-3 hub upgrades.
- Autonomous Bidding Agent for Used-Car Auctions** Apr. 2025  
*Coding Club Recruitment, IIT Guwahati · Python, XGBoost, Pandas, Scikit-learn* Github
  - Trained XGBoost regressor (log-target transform, engineered features: age, usage intensity, condition ratio) on historical US auction data; tuned via RandomizedSearchCV achieving **Val RMSE \$1,620 – 14.3% improvement** over baseline.
  - Designed sigmoid-proportional bidding strategy with bankroll-adaptive ceiling and hard cap at 95% of predicted value, structurally guaranteeing profit on every win; deployed as a live auction simulation agent.
- Bangladesh Power Demand Forecasting** Apr. 2025  
*.ai Club, IIT Guwahati · Python, Scikit-learn, Random Forest, Pandas, NumPy* Github
  - Fused PGCB power records, Dhaka weather observations, and World Bank economic indicators (~92,000 rows, 10-year span) into an end-to-end ML pipeline for hourly electricity demand forecasting.
  - Engineered autoregressive lag features (t-1h, t-24h, t-168h) and cyclical time encodings; tuned Random Forest via GridSearchCV with TimeSeriesSplit CV, achieving **MAPE < 3%** and **R<sup>2</sup> > 0.97** on chronological hold-out.

## TECHNICAL SKILLS

- Programming Languages:** C/C++, Python
- Libraries & Tools:** XGBoost, Scikit-learn, Pandas, NumPy, NetworkX, node2vec, Streamlit, Git, L<sup>A</sup>T<sub>E</sub>X
- Key Concepts:** Machine Learning, Graph Analytics, Time Series Forecasting

## POSITIONS OF RESPONSIBILITY

- Deputy Coordinator**, Coding Club, IIT Guwahati (Machine Learning Module) Jun. 2026 - Present
- Associate**, Finance & Economics Club, IIT Guwahati Jun. 2026 - Present
- Senior Executive**, Entrepreneurship Cell (E-Cell), IIT Guwahati Oct. 2025 - Present
- Associate**, Aeromodelling Club, IIT Guwahati Feb. 2026 - Present
- Associate**, Robotics Club, IIT Guwahati Feb. 2026 - Present

## KEY COURSES TAKEN

- Programming and Data Structures, Linear Algebra, Multivariable Calculus, Real Analysis, Complex Analysis, Machine Learning Specialization (Andrew Ng)

## ACHIEVEMENTS

- Codeforces**, Max Rating 1299 (Handle - om\_sharma\_), 180+ problems solved 2025-Present  
Greedy, Math, BS & DP